How-To Library

Informatica

Creating Multiple Grid Managers on a Hadoop Cluster for Informatica® Big Data Management

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Abstract

When the Blaze engine runs a mapping, it communicates with the Grid Manager, a component that aids in resource allocation, to initialize Blaze engine components on the cluster. You might want to establish two Blaze instances on the same Hadoop cluster. For example, the cluster could host a production instance and a separate instance for testing that does not use production system resources. You can achieve this because each instance uses a separate Grid Manager. You can create a separate Blaze instance by configuring a Data Integration Service to use a unique namespace. This article tells how to create two Blaze instances on a Hadoop cluster.

Supported Versions

- Informatica Big Data Management 10.1.1 Update 2

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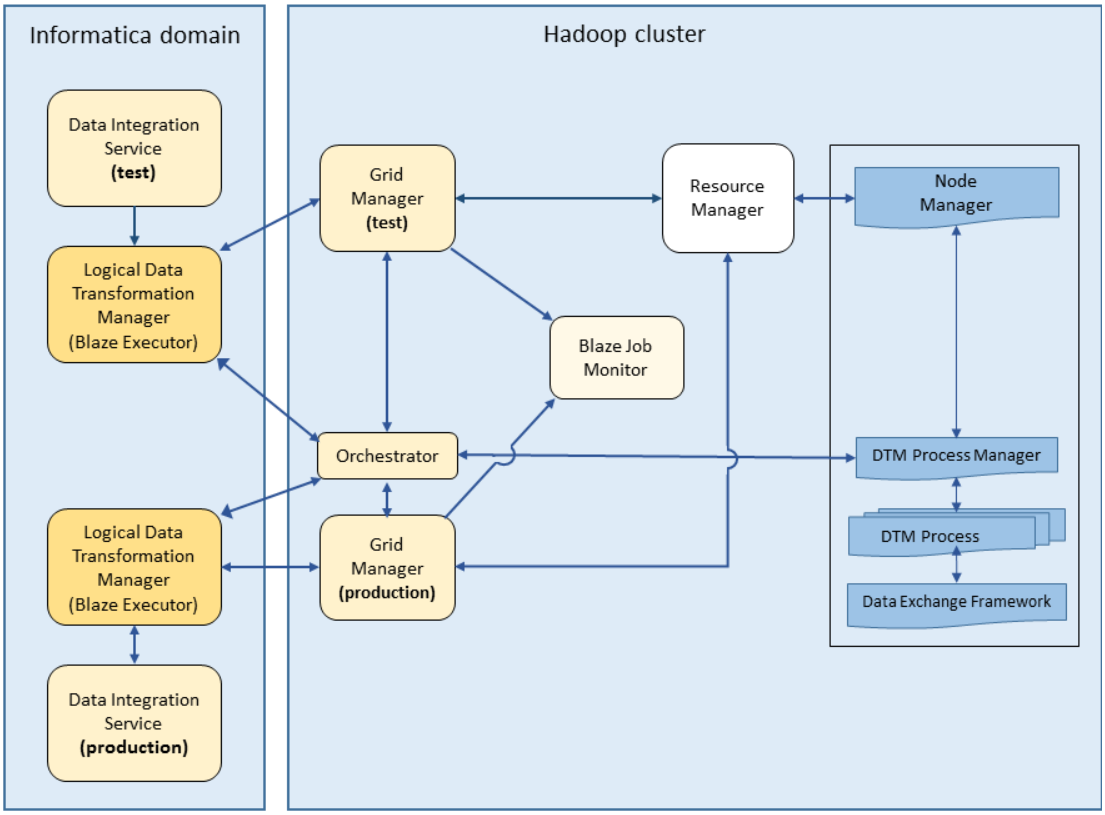
Overview

When you use the Blaze engine to run mappings, Blaze uses a Grid Manager at run time to allot tasks to various nodes in a Hadoop cluster. The Grid Manager aids in resource allocation.

You can use the same Hadoop cluster to stage your test environment and establish a production environment. To control resource use on the cluster, you can establish a separate Blaze instance for testing and another for production.

Each instance requires a separate Grid Manager. You create an additional Grid Manager by performing a series of steps to create separate infrastructure for each Blaze instance, including a unique namespace and a Hadoop connection for each Blaze instance to use.

The following image shows how a separate Data Integration Service on the domain creates a separate Grid Manager on the cluster:



The image shows how separate Data Integration Services use separate Blaze instances. Each instance uses a separate Grid Manager to communicate with the cluster resource manager to balance resources.

Perform the following steps to set up separate Blaze instances:

Step 1. Prepare the Hadoop cluster for the Blaze engine.

Step 2. Configure Data Integration Service properties.

Step 3. Update `hadoopEnv.properties`.

Step 4. Create a new Hadoop connection.

Step 5. Set mapping preferences.

Step 1. Prepare the Hadoop Cluster for the Blaze Engine

To run mappings on the Blaze engine, perform the following tasks:

1. Create an account for the Blaze engine user.
2. Create Blaze engine directories and grant permissions.
3. Grant permissions on the Hive source database.

Create a Blaze User Account

On all nodes in the Hadoop cluster, create an operating system user account for the user you want to run the additional Blaze instance. For example, run the following command:

```
useradd testuser1
```

Create Blaze Engine Directories and Grant Permissions

Create the following directories on the Hadoop cluster:

Local services log directory

Create a local services log directory on all nodes in the cluster and grant permissions to the Blaze user account. The hadoopEnv.properties file on the domain contains an entry for this directory. The file uses an environment variable, $HADOOP_NODE_INFA_HOME, that gets set to the Big Data Management installation directory. The default installation directory is `/opt/Informatica`. For example, run the following commands:

```
hadoop fs mdkir -p /opt/Informatica/blazeLogs
hadoop fs -chmod 777 /opt/Informatica/blazeLogs
```

If you use a different directory name, you must update the following property in the hadoopEnv.properties file: `infagrid.node.local.root.log.dir`

HDFS temporary working directory

Create a working directory on HDFS for the Blaze engine and grant permissions to the Blaze user account. For example, run the following commands:

```
hadoop fs mkdir -p /blaze/workdir
hadoop fs -chmod 777 /blaze/workdir
```

When you configure connection properties, you provide the path to this directory. Alternatively, you can create this directory when you create the connection.

Note: This directory is separate from the aggregated persistent log directory.

Verify that a persistent aggregated HDFS log directory exists on the cluster. For example, `/var/log/Hadoop-yarn/apps/Informatica`.

Note: It is not necessary to create a new directory for persistent logs. Both Blaze instances can use the same persistent aggregated HDFS log directory.

Grant Permissions on the Hive Source Database

Grant the Blaze user account CREATE TABLE permission on the Hive source database. The CREATE TABLE permission is required in the following situations:

- The Hive source table uses SQL standard-based authorization.
- A mapping contains a Lookup transformation with an SQL override.

Step 2. Configure Data Integration Service Properties

Configure Data Integration Service properties to enable two Blaze instances on the Hadoop environment.

You can create a Data Integration Service, or configure one that has not run mappings using the Blaze engine.

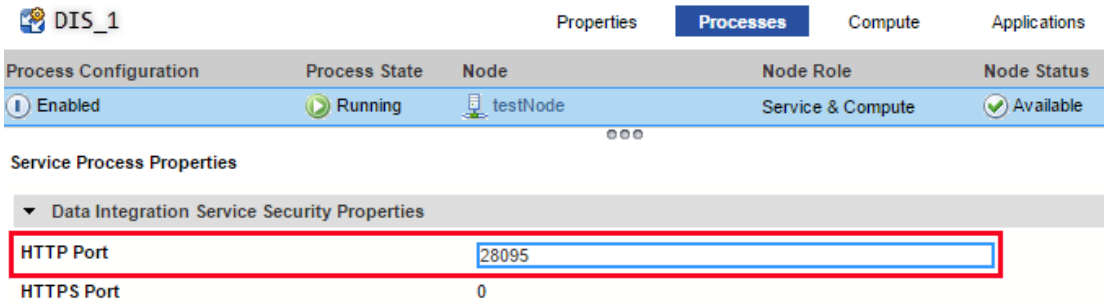
Configure Data Integration Service properties in the Administrator tool.

Data Integration Service Process Properties

Configure the following property on the **Processes** tab:

Property	Description
HTTP Port	The port that the Data Integration Service uses to communicate with the cluster over HTTP. Configure the port with a number that no other process uses.

The following image shows the HTTP Port property:



Data Integration Service Properties

The following table describes the Hadoop properties to configure for the Data Integration Service:

Property	Description
Informatica Home Directory on Hadoop	The Big Data Management home directory on every data node created by the installer. Default is `/opt/Informatica`.
Hadoop Distribution Directory	The directory containing a collection of Hive and Hadoop .jar files on the cluster from the RPM Install locations. The directory contains the minimum set of .jar files required to process Informatica mappings in a Hadoop environment. You can duplicate an existing Hadoop distribution folder if you want to use the same domain as an existing Grid Manager. For example, if you use the distribution folder `cloudera_cdh<version>` for production, you can duplicate it and name the folder like `cloudera_cdh<version>_test`. The distribution folders reside in `<Informatica installation directory>/services/shared/hadoop/`. For example, `/<Informatica installation directory>/services/shared/hadoop/cloudera_cdh<version>`. **Note:** When you use two Informatica domains to run mappings on the same cluster, do not duplicate the Hadoop distribution folder.
Data Integration Service Hadoop Distribution Directory	The Hadoop distribution directory on the Data Integration Service node. Type `/<Informatica installation directory>/Informatica/services/shared/hadoop/<Hadoop distribution name>_<version number>`. For example: `../../services/shared/hadoop/amazon_emr_5.0.0` Configure the Data Integration Service Hadoop Distribution Directory property with the folder on the domain that contains the Hadoop distribution settings that you want to use for the new Blaze instance. **Note:** The contents of the Data Integration Service Hadoop distribution directory must be identical to Hadoop distribution directories on the data nodes.

Step 3. Update hadoopEnv.properties

Update the `hadoopEnv.properties` file on each node where the Data Integration Service runs to configure an additional Blaze instance.

Open `hadoopEnv.properties` and back it up before you configure it. You can find the `hadoopEnv.properties` file in the following location:

```
<Informatica installation directory>/services/shared/hadoop/<distribution name>_<version number>/infaConf
```

Optionally Create a New Namespace

When the machine where the Data Integration Service runs contains two domains running on the same version of Informatica, you configure a new Blaze instance on the domain where you want to run the new Blaze instance.

In the "Advanced Configuration" section of `hadoopEnv.properties`, type the following property to designate a namespace for the Data Integration Service.

infagrid.cadi.namespace

Namespace for the Data Integration Service to use.

Configure the property as follows:

```
infagrid.cadi.namespace=<unique value>
```

For example,

```
infagrid.cadi.namespace=TestUser1_namespace
```

Configure Ports

Search for the following properties and enter port numbers that no other cluster processes use.

infagrid.blaze.console.jsfport

JSF port for the Blaze engine console.

Configure the property as follows:

```
infagrid.blaze.console.jsfport=<unique value>
```

For example,

```
infagrid.blaze.console.jsfport=9090
```

infagrid.blaze.console.httpport

HTTP port for the Blaze engine console.

Configure the property as follows:

```
infagrid.blaze.console.httpport=<unique value>
```

For example,

```
infagrid.blaze.console.httpport=9091
```

Configure Directory Paths

Search for the following properties and enter paths for the Blaze service logs and persistent logs.

infagrid.node.local.root.log.dir

Path for the Blaze service logs.

Note: This is the path that you configured in Step 1 as the local services log directory.

Configure the property as follows:

```
infagrid.node.local.root.log.dir=<directory path>
```

For example,

```
infagrid.node.local.root.log.dir=/opt/Informatica/blazeLogs
```

infacal.hadoop.logs.directory

Path in HDFS for the persistent Blaze logs.

Note: This is the path that you configured in Step 1 as the persistent log directory.

Configure the property as follows:

```
infacal.hadoop.logs.directory=<directory path>
```

For example,

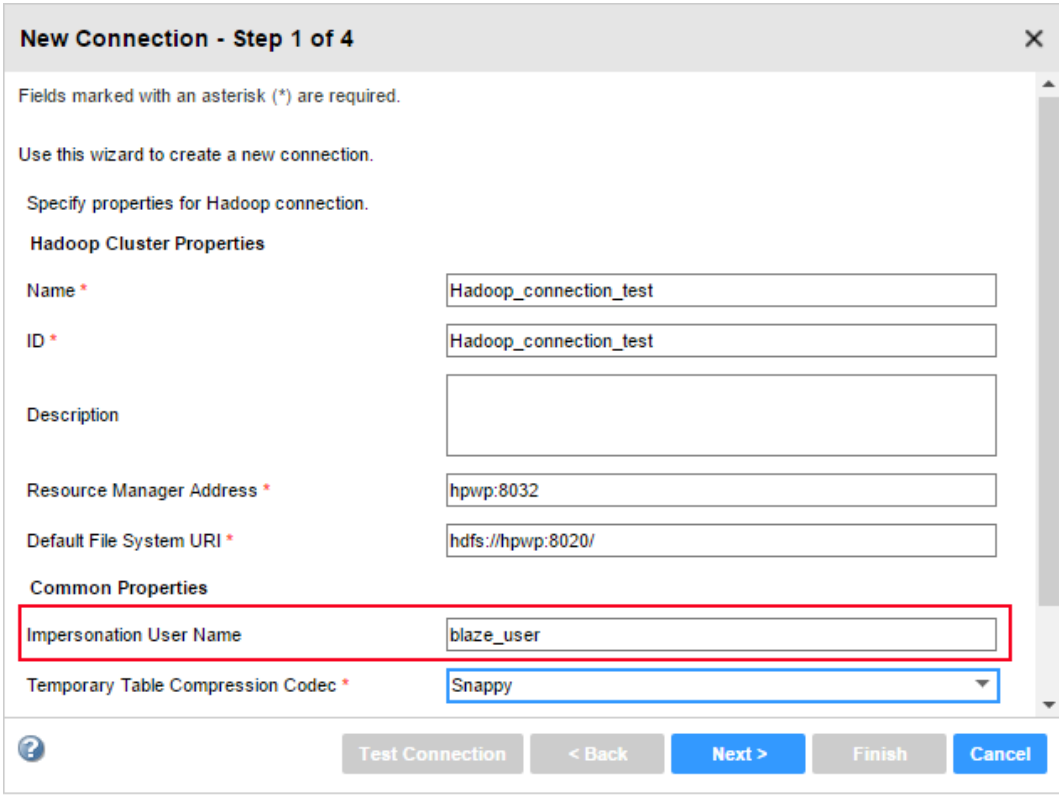
```
infacal.hadoop.logs.directory=infacal.hadoop.logs.directory=/var/log/Hadoop-yarn/apps/
Informatica
```

Step 4. Create a Hadoop Connection

Create a Hadoop connection for the Blaze instance to use.

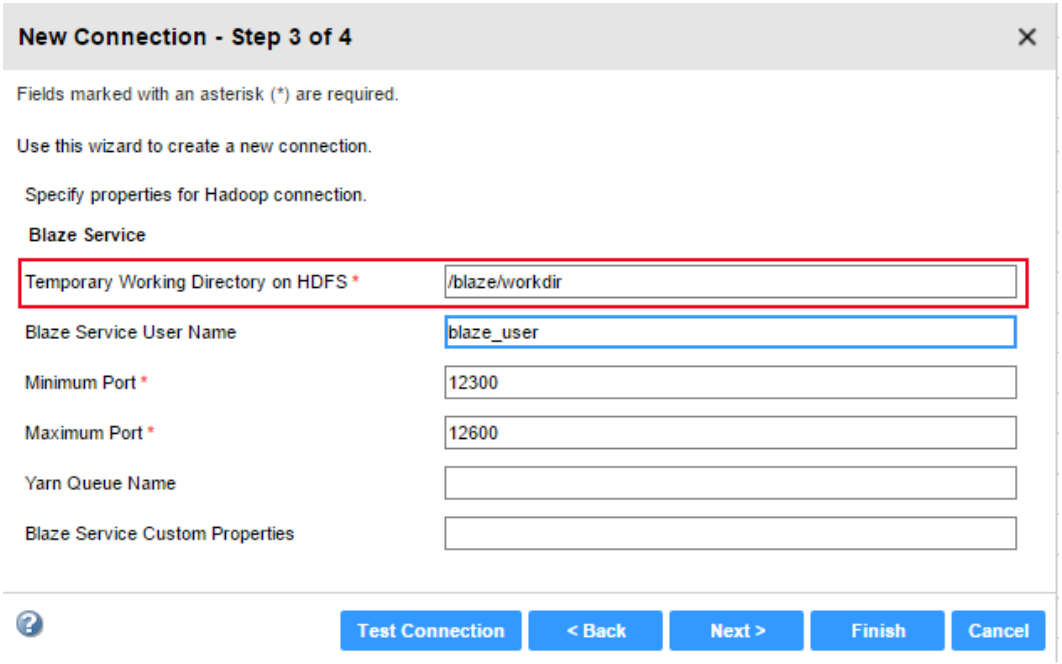
1. In Step 1 of the New Connection wizard, configure the Impersonation User Name property with the same impersonation user that you configured in Step 1, Create a Blaze User Account.

   The following image shows the Impersonation User Name property in the New Connection wizard:



2. In Step 3 of the New Connection wizard, configure the Temporary Working Directory on HDFS property with the path that you configured on the cluster in "Create Blaze Engine Directories and Grant Permissions."

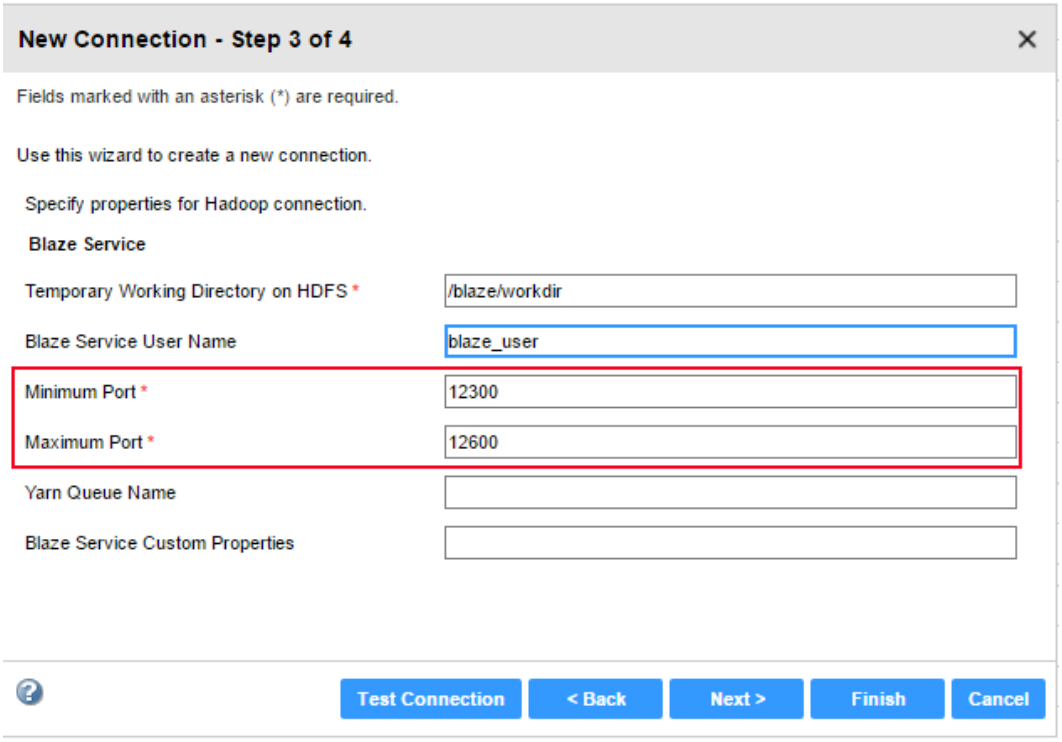
   The following image shows the Temporary Working Directory on HDFS property in the New Connection wizard:



3. Configure the Blaze Service User Name property with the same user name that you used to configure the Impersonation User in Step 1 of this topic.
4. Configure the Minimum Port and Maximum Port properties with a port range for the connection.

   You can supply a range within the port range of an existing Grid Manager, as long as ports are available when the mapping runs. The default range is 300 ports.

   The following image shows the Minimum Port and Maximum Port properties in the New Connection wizard:

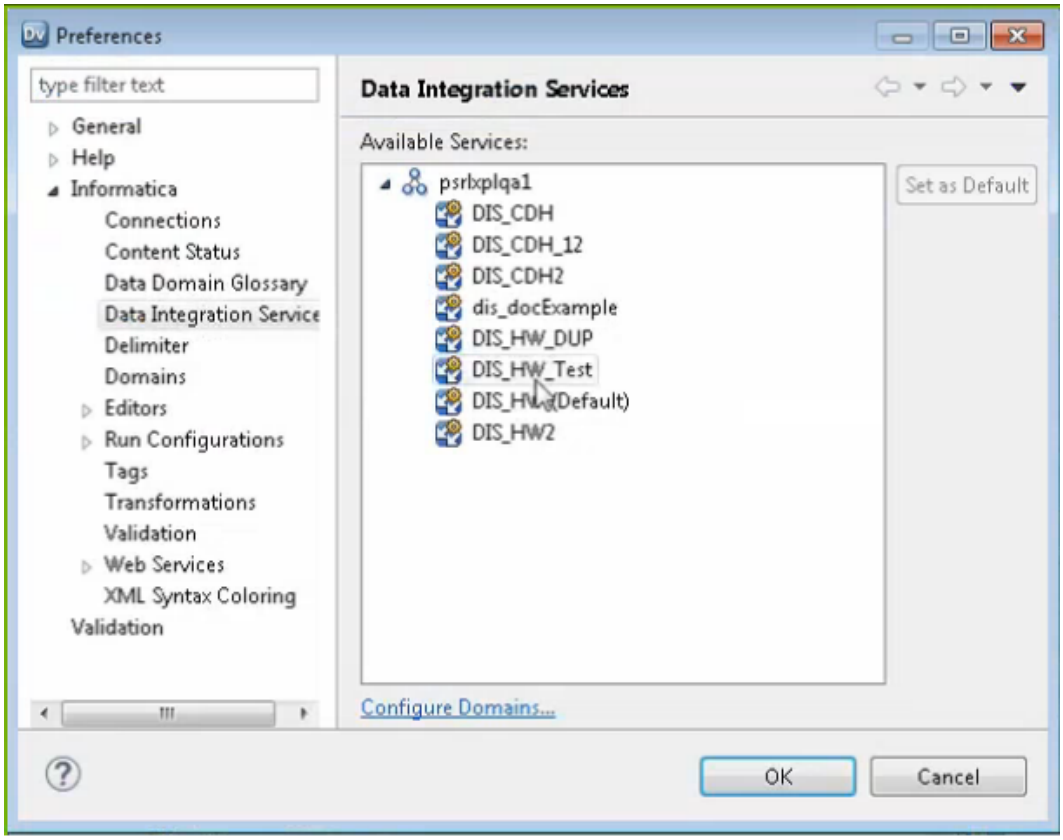


Step 5. Set Mapping Preferences

Before you run the mapping in the Developer tool, configure the mapping to use the Data Integration Service and Hadoop connection you want to use to run the mapping.

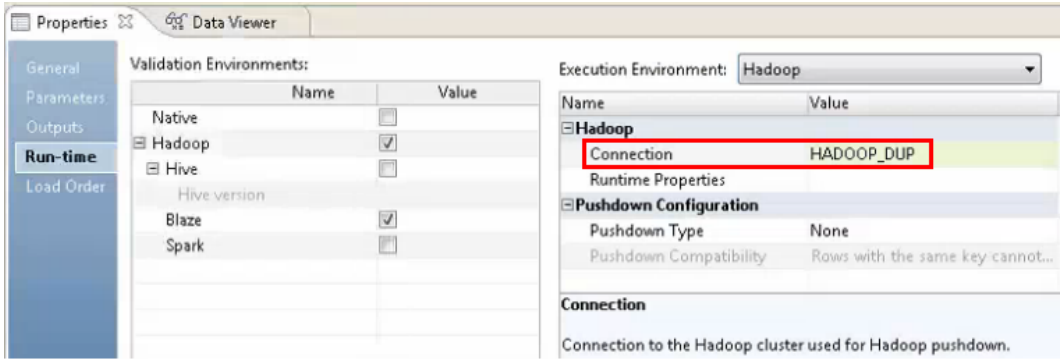
1. In the Developer tool, select **Mapping** > **Preferences**.
2. Expand the **Informatica** node, and then select **Data Integration Service**.

   The following image shows the list of available services in the **Preferences** window:



3. Select the Data Integration Service that you want to use, and then click **OK**.
4. In the **Properties** tab of the mapping, select the **Run-time** sub-tab.
5. In the Execution Environment area, set the Connection property to the Hadoop connection that you created.

   The following image shows the Connection property in the **Properties** tab:

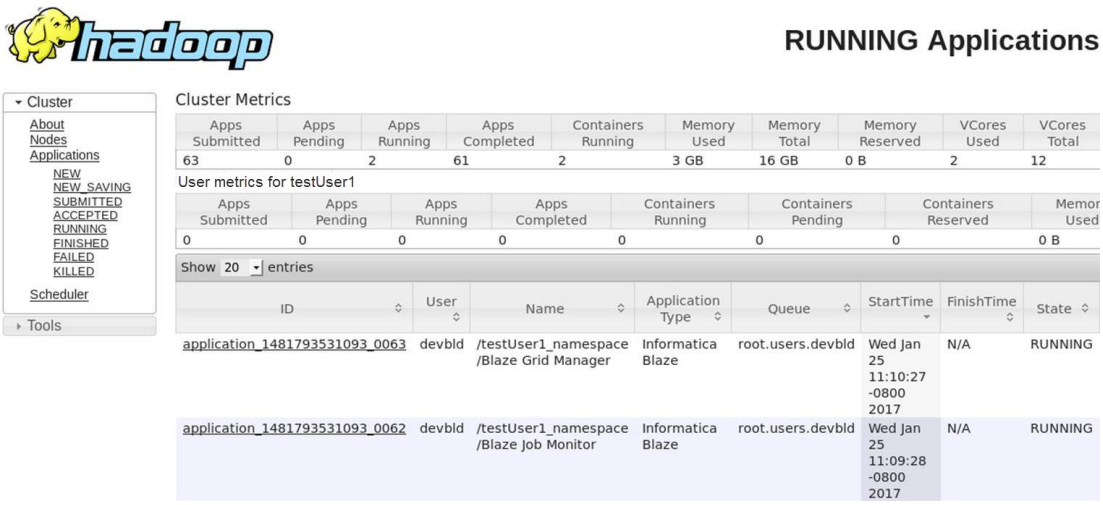


Result

The Data Integration Service creates a Grid Manager on the cluster the first time that it runs a mapping using the Blaze engine.

After you run the mapping, you can verify that the mapping used the Data Integration Service and new Grid Manager that you intended to use to run the mapping. Verify the resources that the mapping used by examining the Running Applications list in the Hadoop Resource Manager web interface. Look for applications that correspond to the namespace that you configured for the Blaze instance.

The following image shows applications with a name that includes the namespace, "testuser1_namespace," that you configured for the Grid Manager:



After completion of the mapping run, the Grid Manager persists. The mapping uses the same Grid Manager whenever it runs with the unique combination of Data Integration Service and connection.

To use multiple connections that use the same Grid Manager, use an identical namespace in each connection to refer to the Blaze instance. Verify that each connection also uses identical values for the Blaze user name and queue name. If you use different values for the Blaze user name and queue name to connect to the same Blaze instance, the mapping fails.

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